Module B3 — Cell Division, Development and Disease

3.1 The Cell Cycle

The cell cycle is the ordered sequence of events that occurs between one cell division and the next. It consists of interphase (the longest phase, where the cell grows and prepares for division) and mitotic phase (where the cell actually divides).

Interphase

Interphase is divided into three stages:

- **G1 (Gap 1):** Cell grows, produces proteins and organelles, carries out normal metabolic functions. The cell increases in size and the number of organelles increases.
- **S (Synthesis):** DNA replication occurs. Each chromosome is copied to produce two identical sister chromatids joined at the centromere. The cell's DNA content doubles from 2n to 4n amount of DNA.
- **G2 (Gap 2):** Further growth and protein synthesis. The cell checks for errors in DNA replication and produces proteins needed for mitosis (e.g., tubulin for spindle fibres). ATP stores are increased.

Cell cycle: The regulated sequence of events between one cell division and the next, consisting of interphase (G1, S, G2) and mitotic phase (mitosis and cytokinesis).
Checkpoints: Control points in the cell cycle where the cell assesses whether conditions are suitable to proceed. Key checkpoints exist at G1/S boundary, G2/M boundary, and during mitosis.

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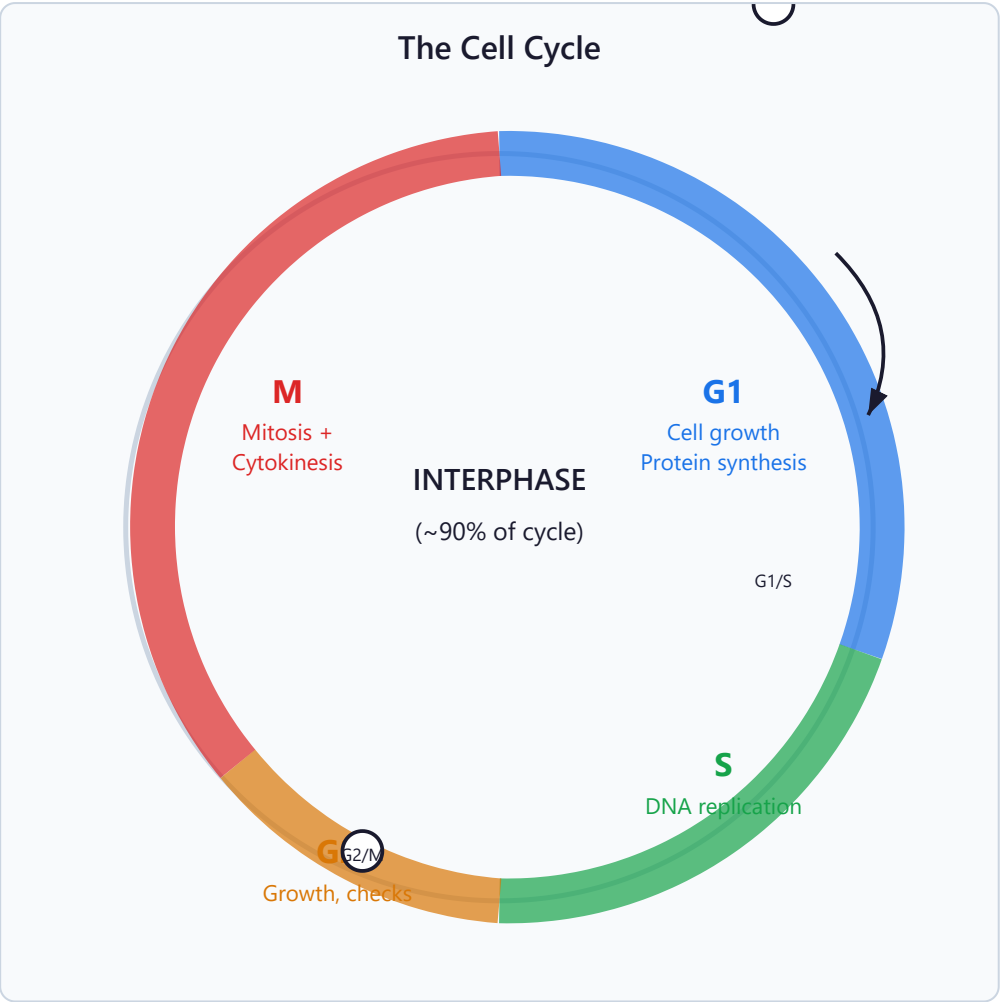

Mitosis

Mitosis produces two genetically identical daughter cells from one parent cell. It is used for growth, repair, and asexual reproduction. Mitosis consists of four continuous stages:

1. **Prophase:** Chromatin condenses into visible chromosomes (each consisting of two sister chromatids). The nuclear envelope begins to break down. Centrioles move to opposite poles and spindle fibres begin to form.
2. **Metaphase:** Chromosomes line up along the cell equator (metaphase plate). Spindle fibres attach to centromeres of each chromosome.
3. **Anaphase:** Centromeres divide and sister chromatids are pulled to opposite poles by shortening of spindle fibres. The cell elongates.
4. **Telophase:** Chromatids arrive at opposite poles and decondense back into chromatin. Nuclear envelopes reform around each set of chromosomes. Nucleoli reappear.

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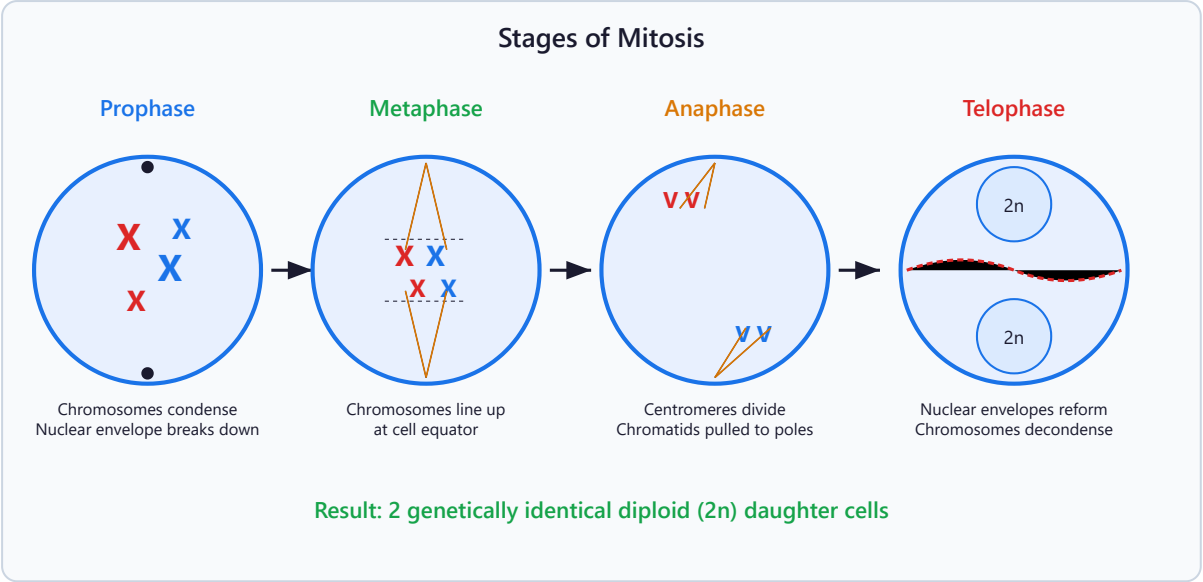


Cytokinesis then divides the cytoplasm. In animal cells, a cleavage furrow forms and pinches the cell in two. In plant cells, a cell plate forms from vesicles along the equator, which develops into a new cell wall.

> **Exam Tip:** Remember that DNA replication occurs during S phase of interphase, NOT during mitosis. By the time mitosis begins, each chromosome already consists of two identical sister chromatids. A common exam mistake is saying "chromosomes replicate during prophase" — this is incorrect.

Significance of Mitosis

- **Growth:** Increasing cell number in multicellular organisms during development
- **Repair:** Replacing damaged or dead cells (e.g., wound healing, blood cell replacement)
- **Asexual reproduction:** Producing genetically identical offspring (e.g., binary fission in bacteria uses a different mechanism, but vegetative propagation in plants involves mitosis)
- **Genetic consistency:** All daughter cells are genetically identical to the parent cell, maintaining chromosome number and genetic stability

Cancer: Uncontrolled Cell Division

Cancer results from mutations in genes controlling the cell cycle. Two key gene types are involved:

- **Proto-oncogenes** normally stimulate cell division. When mutated into **oncogenes**, they become permanently active, driving excessive cell division (like a stuck accelerator pedal).
- **Tumour suppressor genes** (e.g., p53, Rb) normally inhibit cell division or promote apoptosis. When inactivated by mutation, the cell cycle proceeds unchecked (like broken brakes).

Typically, multiple mutations in several genes are required before a cell becomes cancerous (the multi-hit hypothesis). Risk factors include UV radiation, ionising radiation, carcinogenic chemicals (e.g., in tobacco smoke), certain viruses (e.g., HPV causes cervical cancer), and inherited genetic predispositions (e.g., BRCA1/2 mutations increase breast cancer risk).

Benign tumours are localised, enclosed in a capsule, grow slowly, and do not invade surrounding tissues or spread. **Malignant tumours** (cancers) invade surrounding tissues and can metastasise (spread to distant sites via the blood or lymphatic system), forming secondary tumours.

3.2 Meiosis

Meiosis is a reduction division that produces four genetically different haploid daughter cells from one diploid parent cell. It occurs in the gonads (ovaries and testes) to produce gametes for sexual reproduction.

Meiosis I (Reduction Division)

- **Prophase I:** Homologous chromosomes pair up (bivalents). Crossing over occurs between non-sister chromatids at points called chiasmata, exchanging genetic material. Chromosomes condense, nuclear envelope breaks down.
- **Metaphase I:** Bivalents line up randomly at the equator (independent assortment). Each pair can face either way, giving 2^n possible arrangements (where n = haploid number).
- **Anaphase I:** Homologous chromosomes are pulled to opposite poles (whole chromosomes, not chromatids). This halves the chromosome number.
- **Telophase I:** Nuclear envelopes may reform. Cytokinesis produces two haploid cells, each with one chromosome from each homologous pair.

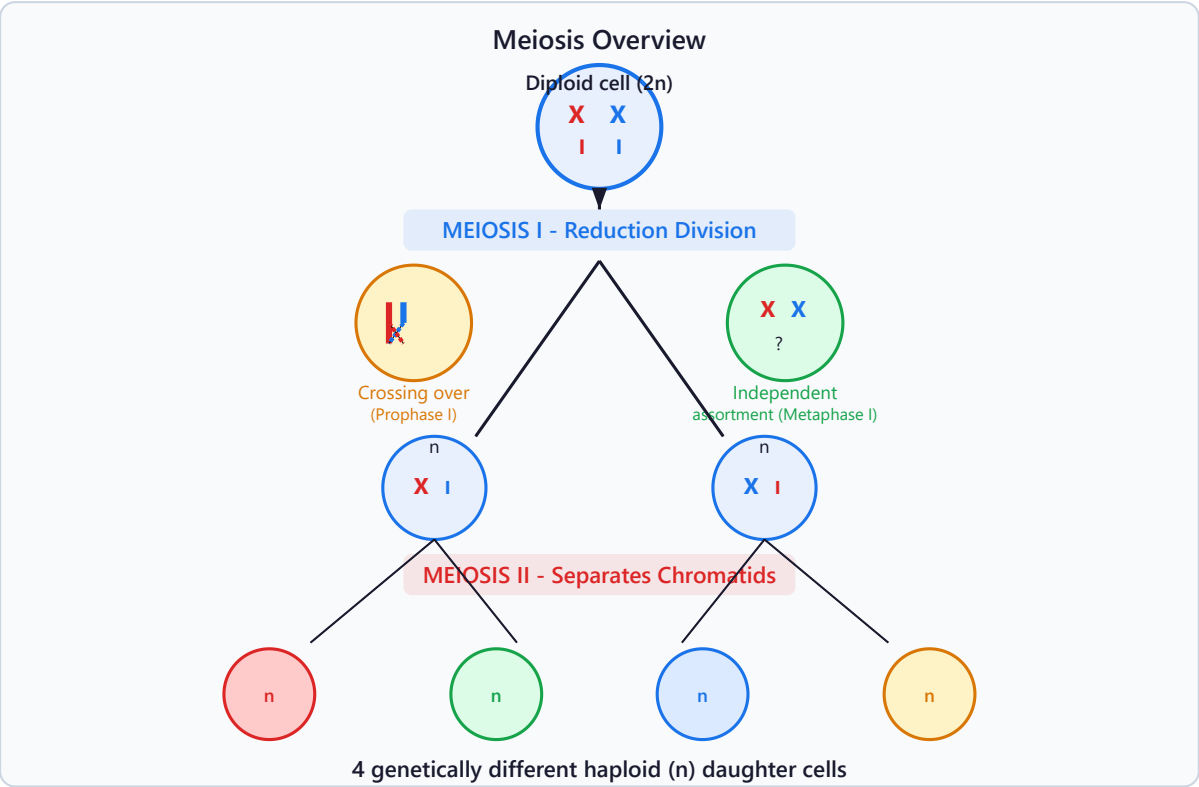
Meiosis II

Meiosis II is essentially identical to mitosis. Sister chromatids are separated, producing four haploid cells that are all genetically different from each other and from the parent cell.

Sources of Genetic Variation

- **Crossing over:** Exchange of alleles between homologous chromosomes during prophase I creates new combinations of alleles on each chromosome

- **Independent assortment:** Random orientation of bivalents at metaphase I means each gamete receives a random combination of maternal and paternal chromosomes
- **Random fertilisation:** Any male gamete can fuse with any female gamete, further increasing variation



Haploid (n): A cell containing one set of chromosomes (e.g., human gametes have 23 chromosomes).
Diploid (2n): A cell containing two sets of chromosomes, one from each parent (e.g., human body cells have 46 chromosomes).

3.3 Stem Cells and Differentiation

Types of Stem Cell

Stem cells are undifferentiated cells capable of dividing by mitosis and differentiating into specialised cell types. They vary in their potency:

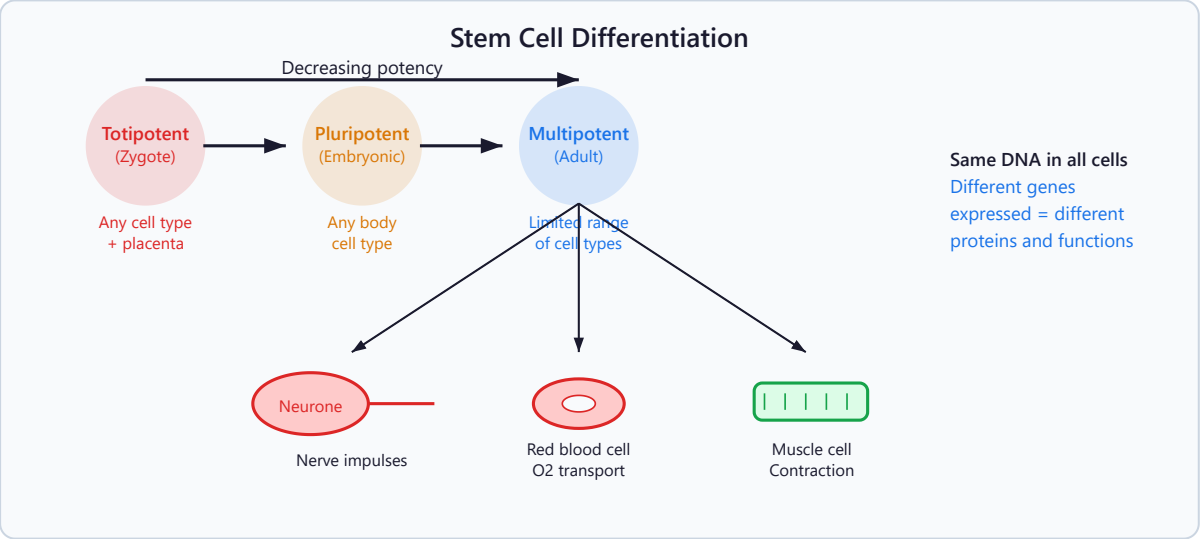
- **Totipotent:** Can differentiate into any cell type including extraembryonic tissue (e.g., zygote and cells up to about the 16-cell stage)

- **Pluripotent:** Can differentiate into any cell type of the body but not extraembryonic tissue (e.g., embryonic stem cells from the inner cell mass of a blastocyst)
- **Multipotent:** Can differentiate into a limited range of related cell types (e.g., adult haematopoietic stem cells can produce all blood cell types)
- **Unipotent:** Can only produce one cell type but retain the ability to self-renew (e.g., skin stem cells producing keratinocytes)

Differentiation

Differentiation is the process by which a cell becomes specialised for a particular function. It involves the selective expression of certain genes while others are switched off. All cells contain the same DNA, but different genes are expressed in different cell types, producing different proteins and therefore different structures and functions.

Gene expression is regulated by transcription factors, epigenetic modifications (e.g., DNA methylation, histone modification), and signalling molecules from neighbouring cells during development.



Medical Applications of Stem Cells

Stem cells have potential therapeutic applications including treatment of leukaemia (bone marrow transplants), repair of damaged tissues (e.g., cardiac muscle after heart attack), and potentially treating neurodegenerative diseases. Ethical concerns include the destruction of embryos to obtain embryonic stem cells, though induced pluripotent stem cells (iPSCs) made by reprogramming adult cells offer an alternative.

3.4 The Immune System

Non-Specific Defences

The body's first line of defence is non-specific (does not distinguish between different pathogens):

- **Skin:** Physical barrier; outer layer of dead keratinised cells; sebaceous glands produce antimicrobial sebum
- **Mucous membranes:** Mucus traps pathogens; cilia waft mucus away from lungs
- **Stomach acid:** Low pH (approximately 2) kills most ingested microorganisms
- **Lysozyme:** Enzyme in tears, saliva, and nasal secretions that destroys bacterial cell walls
- **Inflammation:** Damaged cells release histamine, causing vasodilation and increased capillary permeability, bringing more white blood cells to the site of infection
- **Phagocytosis:** Neutrophils and macrophages engulf and digest pathogens in vesicles using lysosomes

Specific Immune Response

The specific (adaptive) immune response is targeted against particular pathogens and involves lymphocytes recognising specific antigens.

> **Antigen:** A molecule (usually a protein or glycoprotein) on the surface of a cell that can trigger an immune response. Self-antigens are recognised as belonging to the body; non-self antigens are foreign.
> **Antibody:** A Y-shaped glycoprotein (immunoglobulin) produced by plasma cells. Each antibody has a specific variable region that is complementary to one particular antigen.

Cell-Mediated Immunity (T Cells)

T lymphocytes mature in the thymus and respond to antigens presented on the surface of infected cells by antigen-presenting cells (APCs). Types include:

- **T helper cells (CD4+):** Release cytokines that activate B cells and cytotoxic T cells
- **Cytotoxic T cells (CD8+):** Kill infected body cells by releasing perforin (forms pores in the target cell membrane) and granzymes
- **T memory cells:** Remain in the body long-term for a faster secondary response

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Humoral Immunity (B Cells)

B lymphocytes mature in the bone marrow. When activated (by T helper cells and antigen binding), they undergo clonal expansion and differentiate into:

- **Plasma cells:** Short-lived cells that produce and secrete large quantities of specific antibodies
- **B memory cells:** Long-lived cells that remain in the blood and lymph, enabling a rapid secondary immune response upon re-exposure to the same antigen

Antibodies destroy pathogens by: agglutination (clumping pathogens together), neutralisation (blocking pathogen attachment to host cells), opsonisation (coating pathogens to enhance phagocytosis), and activating complement proteins.

Primary and Secondary Immune Response

The primary response is slow (takes days to weeks) because there are few lymphocytes specific to the new antigen. The secondary response is much faster and stronger because memory cells are already present in large numbers and can rapidly divide and differentiate upon re-exposure.

3.5 Vaccination

Vaccination: The deliberate introduction of antigenic material (which is not pathogenic) into the body to stimulate an immune response and the production of memory cells, providing immunity without causing disease.

Vaccines may contain attenuated (weakened) pathogens, inactivated (killed) pathogens, isolated antigens (subunit vaccines), or toxoids (inactivated toxins). The key principle is that the immune system is exposed to antigens from the pathogen without the person becoming ill.

Herd immunity occurs when a sufficiently high proportion of a population is vaccinated, reducing transmission to a level where even unvaccinated individuals are unlikely to encounter the pathogen. The threshold for herd immunity varies by disease (typically 85-95%).

Exam Tip: Vaccines do not contain antibodies. They contain antigens that stimulate the body's own immune system to produce antibodies and, crucially, memory cells. This is active immunity because the body produces its own immune response.

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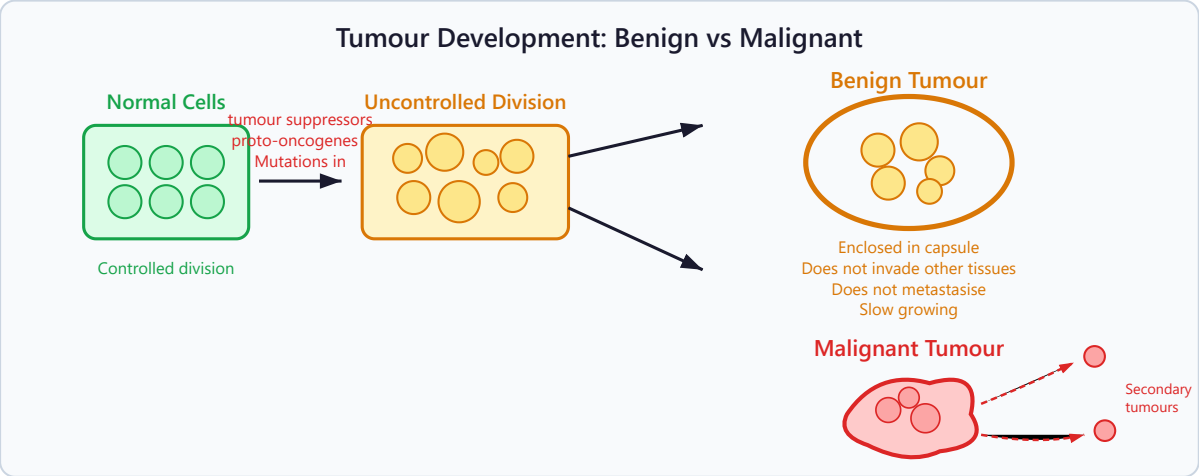
3.6 Communicable Diseases and Pathogens

Types of Pathogen

Pathogen Type	Examples	Characteristics
Bacteria	Tuberculosis (Mycobacterium tuberculosis), cholera (Vibrio cholerae)	Prokaryotic; cause disease by producing toxins or destroying tissue; treatable with antibiotics
Viruses	HIV, influenza, SARS-CoV-2	Non-living; obligate intracellular parasites; replicate using host cell machinery; not affected by antibiotics
Fungi	Athlete's foot (Trichophyton), ringworm	Eukaryotic; may produce spores; treated with antifungals
Protoctists	Malaria (Plasmodium), sleeping sickness (Trypanosoma)	Eukaryotic; often complex life cycles involving vectors

Transmission

Communicable diseases can be transmitted by direct contact, airborne droplets, contaminated water or food, vectors (organisms that carry pathogens between hosts, e.g., mosquitoes for malaria), or bodily fluids.



Antibiotics and Resistance

Antibiotics kill or inhibit bacteria (e.g., penicillin inhibits cell wall synthesis). They are ineffective against viruses. Antibiotic resistance arises through natural selection: random mutations may confer resistance; in the presence of antibiotics, resistant bacteria survive and reproduce, passing on resistance genes. Overuse and misuse of antibiotics accelerates this process.

MRSA (methicillin-resistant Staphylococcus aureus) is a well-known example of antibiotic-resistant bacteria. Strategies to combat resistance include completing full courses of antibiotics, restricting unnecessary prescribing, developing new antibiotics, and improving hygiene to reduce infection spread.

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